

-----BEGIN PGP SIGNED MESSAGE-----

Who controls the way out?

If every path out of a shared UTXO either pays you or can be countered by you, you keep Bitcoin-level custody.

// Bitcoin Layer 2 custody · theGig · 10/10/26

```
020000000000101f3a9c2e17b4d80a5
6e2c91d04b7f3a88e15c6d2a9b0f47e3
00000000fdffffff02a08601000000
0000225120c47e9a1b3d5f80e26a71
9c4b0de38f52a617bc09e4d1f68a2b
3c75e0914d6af28b17e3c50a9d4f61
0140a1f3e9c47b2d058e6a91c3f07b
4d2e85a61c9f30b7e4d18a5c2f6e09
```

// BITCOIN TODAY

On Bitcoin, what you own is a UTXO

1 · You hold the key

Your key is the only thing that can authorize spending this coin.

2 · The path is sound

The rules for spending it are enforced by cryptography, not by a company.

3 · It lives on the blockchain

The coin is recorded on Bitcoin itself, checked by every node.

4 · It is settled

Everyone agrees it is provably yours. Nothing is pending.

// BITCOIN TODAY

Why we trust it



Math

A signature can't be forged without the key.



Many verifiers

Thousands of independent nodes reject any invalid spend.



Time

Each new block makes the record harder to rewrite.

Security is passive. Do nothing for ten years, and the coin is still yours.

Your only job is to keep the key safe.

// THE STEP TO LAYER 2

**Block space is scarce,
so many people share one UTXO.**

Lightning, Ark and other Layer 2s all start from the same idea: one real coin on Bitcoin, divided among many owners off-chain.

// SHARED UTXO

One real coin, many virtual ones

ON BITCOIN · SETTLED

One shared UTXO, locked by several keys



Your share

virtual UTXO

Alice's share

virtual UTXO

Bob's share

virtual UTXO

...and more

virtual UTXO

Each share is backed by **already-signed transactions** its owner holds, ready to broadcast.

// UTXO VS. VIRTUAL UTXO

Same four statements, one changes

Statement	Bitcoin UTXO	Virtual UTXO (e.g. Lightning)
You hold a key that controls the path	Yes	Yes, plus a signed closing transaction
The path is cryptographically sound	Yes	Yes, but other valid paths also exist
It is grounded on the blockchain	Yes	Yes, through the shared UTXO
Settled and provably yours	Yes	Yours, conditionally. Not yet settled.

// WHAT CARRIES OVER

Same custody, different security

Custody: the same

- No one can move your share without your signature, or your failure to respond
- You can leave alone, without asking permission
- The exit lands as an ordinary Bitcoin coin

Security: active, not passive

- **Time:** you may need to act within a window
- **Cost:** exiting costs on-chain fees
- **Records:** you must keep the latest signed state

Think of it as an **unexercised option**: yours to claim, as long as you can claim it in time and it's worth claiming.

// THE DEFINITION

A virtual UTX0 is provably yours if every valid path out of the shared UTX0 either pays you, or can be countered by you: alone, in time, for less than it's worth.

Pays you

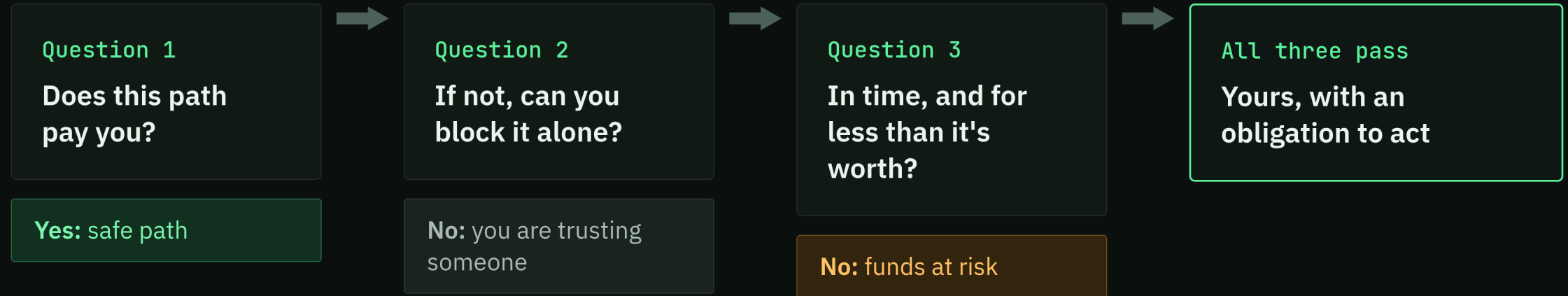
Alone

In time

Worth it

// APPLYING THE DEFINITION

Three questions for every path



Run every path through this. One failing path is enough to mean you're trusting someone with your funds.

// THE PATH ANALYSIS TOOL

Map the paths, test them, measure the risk

01

Map

List every valid way the shared UTXO can be spent, and whose keys each one needs.

02

Test

Check each path against the definition: pays you, or you can counter it alone, in time, for less than it's worth.

03

Measure

Estimate how likely you are to miss a counter: being offline, fee spikes, lost keys or records.

The result: a map of every way out, a pass or fail for each, and a risk estimate with its assumptions visible.

// EXAMPLE 1 OF 5 • ON-CHAIN

A single-key Bitcoin UTXO

\$ PATHS --OUT UTXO

You sign with your key

Pays you

Anyone else

No valid path

> VERDICT

**Provably yours,
passively**

Your obligation

Keep the key safe and backed up.

Main risk driver

Losing the key, or having it stolen. One point of failure.

// EXAMPLE 2 OF 5 • ON-CHAIN

Multisig: 2-of-3 and 4-of-5

\$ PATHS --OUT UTXO

You sign with enough keys (2 of 3, or 4 of 5)

Pays you

A co-signer or custodian holding one key

Can't spend alone

A thief gets enough keys

Theft: 4-of-5 is
much harder

You lose two keys

Locked out in both setups

> VERDICT

**Provably yours,
while you control
enough keys**

Your obligation

Keep keys in separate places, plus a backup of the wallet setup.

Main risk driver

Keys that share a place or a person.
One fire or break-in can hit several.

// EXAMPLE 3 OF 5 • LAYER 2

A Lightning channel

\$ PATHS --OUT SHARED-UTXO

You broadcast your latest closing transaction

Pays you, after a delay

Your peer broadcasts their latest one

Pays you immediately

You both agree to close

Needs your signature

Your peer broadcasts an old, outdated state

You counter: a penalty
takes it all, within
the window

> VERDICT

**Yours, with an
obligation to watch**

Your obligation

Be online within the window,
or use a watchtower. Keep your
channel backup.

Main risk driver

Being offline when cheated, or fees
too high for a small channel.

// EXAMPLE 4 OF 5 • LAYER 2

An Ark virtual UTXO (VTXO)

\$ PATHS --OUT SHARED-UTXO

You broadcast your branch of signed transactions

Pays you, after a delay

You refresh or pay with the operator

Needs your signature

The operator refuses to serve you

You counter: exit alone

The operator sweeps the shared UTXO after it expires

You counter: refresh or exit before expiry

> VERDICT

Yours, with a scheduled deadline

Your obligation

Refresh before expiry. Your wallet or a delegate can do it for you.

Main risk driver

Going dormant past expiry, or a VTXO too small to be worth exiting.

// EXAMPLE 5 OF 5 • NEW PROPOSAL (2026)

Shielded Bitcoin: a private note

\$ PATHS --OUT SHIELDED-NOTE

You spend it with your wallet's spending key

Only you can authorize

An indexer hides or delays your data

You counter: rebuild state
from Bitcoin yourself

Someone forges a spend proof

Blocked if the proof setup
was honest

You exit to an ordinary Bitcoin UTXO (peg-out)

Planned via PIPEs v2:
next slide

> VERDICT

**Yours inside the
system. The way out is
planned, not yet
specified.**

Your obligation

Guard your wallet seed. Check the
chain yourself, or accept trusting
an indexer.

Main risk driver

The peg-out design, which decides
whether you can leave alone.

// EXAMPLE 5, CONTINUED · PIPES V2 · RESEARCH STAGE

The exit: a key unlocked by proof

The peg-out UTXO is an ordinary address. Its key is encrypted so only a valid proof can unlock it.

\$ PATHS --OUT PEG-OUT-UTXO

You prove your peg-out, unlock the key, sign

Pays you: no deadline,
no operator

Setup participants kept a copy of the key

Assumption: honest multi-
party setup

Someone breaks the encryption scheme

Assumption: security not
yet proven

Someone else holds a valid proof for your exit

Race: the proof must be tied
to you

No valid proof can be produced

Frozen: rebuild from Bitcoin
with your seed

> VERDICT

**Unilateral and
passive, if the
assumptions hold**

Your obligation

Guard your seed. Nothing to watch,
no deadline.

Main risk driver

Trust in the setup and a new
encryption scheme. Not yet practical
to deploy.

// SIDE BY SIDE

Five systems, one test

System	Who else has a path?	Your counter	Your obligation	Verdict
Single key	No one	None needed	Guard the key	Yours, passive
Multisig	Co-signers, only together	Hold enough keys	Keep keys apart	Yours, passive
Lightning	Your peer	Penalty transaction	Watch within the window	Yours, active
Ark	Operator, after expiry	Refresh or exit	Act before the deadline	Yours, active
Shielded Bitcoin	No one, if setup and encryption hold	None needed (PIPE exit)	Guard the seed	Yours, on assumptions; research stage

If you're comfortable with the paths out, you can be comfortable with the Layer 2.

Custody carries over

No one moves your share without you, or your failure to respond.

Security becomes active

Time, fees and records are the new conditions.

The risk is measurable

Map the paths, test them, and put numbers on what you must do.

-----END PGP SIGNATURE-----

<https://claude.ai/artifact/QwdZ45jfsJJTPYThAcApux>